

Comparing a Yoga Class with a Resistance Exercise Class: Effects on Body Satisfaction and Social Physique Anxiety in University Women

Kimberley L. Gammage, Breanne Drouin, and Larkin Lamarche

Purpose: The current study compared a single yoga group exercise class and a resistance group exercise class for their effects on state body satisfaction and social physique anxiety in women. **Methods:** A pretest-posttest design was used. Participants ($N = 46$) completed both a resistance exercise class and yoga class in a counterbalanced order. Measures of body satisfaction and social physique anxiety were completed immediately before and after each class. **Results:** A 2 (time) $\times$ 2 (class type) repeated-measures multiple analysis of variance showed a significant overall Time $\times$ Class Type interaction ($F_{2,44} = 5.69, P < .01, \eta_p^2 = .21$). There was a significant increase in body satisfaction after the yoga class. After both classes, there was a significant decrease in social physique anxiety, but the magnitude of the change was larger after the yoga class than after the resistance class. **Conclusions:** Both types of exercise class were associated with improvements in body image, but there were greater improvements after the yoga class. This study provided evidence of the positive effects of yoga for reducing state social physique anxiety and increasing state body satisfaction, adding to correlational evidence suggesting that yoga is particularly beneficial for improving body image-related outcomes in women.

Keywords: body image, physical activity, acute exercise, group exercise

Body image refers to an individual's thoughts, feelings and beliefs about his or her body in terms of appearance and overall functionality.¹ Negative body image is a risk factor for a number of behaviors and emotional and mental disorders that could lead to poor health outcomes, such as excessive dieting, physical inactivity, excessive exercise, disordered eating, negative affect, depression, and decreased self-esteem.¹ One approach to improving body image is exercise. Three meta-analyses have reported that exercise is associated with small to moderately sized improvements in body image for men and women across the lifespan.²⁻⁴ Further, these meta-analyses have generally shown that although exercise intensity moderated this effect, characteristics such as duration, intervention length, and mode of exercise (aerobic, resistance, or a combination of the 2) did not.^{2,3}

However, there are 2 important limitations to the extant research. First, although the meta-analyses found that exercise mode was not a significant moderator of the relationship between exercise and body image, only traditional types of fitness activities were examined (ie, walking, aerobic dance, cycling, strength training). People participate in many other types of fitness activities today, such as yoga. Yoga is increasing in popularity in North America. For example, it was estimated that in 2016 over 36.7 million Americans practice yoga, which represents a growth of 50% since 2012.⁵ Yoga is promoted as an activity that can improve strength, flexibility, and balance but is equally beneficial for stress relief, self-acceptance, mood improvements, and internal body awareness.⁶ In fact, flexibility and stress relief are the 2 most commonly cited reasons for beginning and continuing yoga practice.⁵ However, despite the

popularity and health benefits of yoga, there has been limited experimental research on the potential benefits for body image outcomes, such as social physique anxiety or body satisfaction.

Correlational research has suggested that yoga may be associated with more positive body image compared with other types of exercise.⁷⁻¹⁰ For example, in one study, yoga practitioners reported greater body awareness (paying attention to the body's physical and affective internal sensations) and body responsiveness (using the body's internal sensations to guide reactions and movements), higher body satisfaction and lower self-objectification than aerobic exercisers or nonexercisers.⁷ A second study similarly showed that the time spent in aerobic activities was positively correlated with self-objectification and eating disorder symptoms, and negatively associated with body esteem, whereas participation in yoga was negatively associated with self-objectification.¹⁰ They also found that time spent weight training was unrelated to any body image variable. However, because these studies were correlational, a causal relationship cannot be determined.

There is also evidence that yoga can positively affect body image outcomes in an eating disordered population or in those with negative body image.^{11,12} In nonclinical samples, yoga is associated with improvements in body image in adolescent girls.^{13,14} However, because yoga was only one part of a multimodal intervention, its effects cannot be distinguished from other components of the program.

The second limitation to existing research is that relatively few studies have investigated the effect of a single bout of exercise on state body image. Presently, there is limited evidence to suggest that a single session of exercise can affect state body image.¹⁵⁻¹⁷ For instance, Lamarche et al¹⁵ found that completion of a step aerobics class led to a decrease in state social physique anxiety, regardless of the presence or absence of mirrors. Likewise, female participants reported lower social physique anxiety after a resistance exercise

The authors are with the Dept of Kinesiology, Brock University, St. Catharines, Ontario, Canada. Gammage (kgammage@brocku.ca) is corresponding author.

class, regardless of the gender of the exercise leader.¹⁶ These studies provide evidence that a single exercise session can positively affect state body image.

Examining the acute effects of exercise on body image may have important practical implications. Although a single positive experience may increase one's intention to participate in future exercise, a single negative experience may act as a deterrent to future participation. For instance, Dishman has noted that enjoyment and well-being are important factors in exercise adherence and are probably more important than factors such as health benefits^{18,19} Further, it has been noted that feelings of pleasure or displeasure experienced during an exercise session can affect future decisions to either adhere to exercise or drop out.²⁰ Although evidence suggests that a single exercise session can change state body image,^{15–17} we currently do not know whether this finding applies equally to all modes of exercise. To date, research has investigated only traditional fitness activities (ie, aerobics, resistance training). Practitioners need to be aware of the importance mode of exercise may have on future exercise participation. For example, not all modes of exercise may be equally effective on all outcomes; the emphasis on the mind-body link of yoga may make it particularly useful for improving body image outcomes, which in turn could improve adherence.

The purpose of the current study was to address these 2 major limitations in the exercise and body image literature. We examined the effects of a yoga group exercise class on state body image in college women compared with the effects of a resistance exercise class, which has been shown previously to reduce social physique anxiety.¹⁵ It was hypothesized that after both types of classes, state body satisfaction would increase whereas state social physique anxiety would decrease, but that larger changes would be seen after the yoga class than after the resistance class.

Methods

Participants

Upon institutional ethics clearance, participants were recruited from a university campus in southern Ontario. College women were recruited because body image concerns are particularly relevant for women, who experience greater body dissatisfaction than men do.^{21,22} Further, body image outcomes such as social physique anxiety are typically higher in women than in men.^{23,24} In addition, from a practical perspective, because the majority of people who participate in group fitness classes are women, ecological validity was enhanced by recruiting only women. Individuals who exercised less than twice per week (nonexercisers or infrequent exercisers) were recruited to eliminate potential effects that exercise experience may have on study variables^{25–27} and because those who exercise less than twice per week differ on a variety of exercise-related variables (eg, exercise imagery, self-efficacy^{25,26,28}) from those who exercise 3 or more times per week. Further, research has consistently shown more negative body image in those who exercise less, including both body satisfaction and social physique anxiety.^{2–4,26} Finally, individuals who already exercise may experience a ceiling effect, such that a single exercise class may not be able to further improve their body image.²⁹ Participants were excluded if they were physically unable to participate in the exercise classes, were varsity athletes, or had a history of a clinical eating disorder.

A total of 55 participants contacted the research team to participate. Five participants were excluded because they exercised 3 or more times per week. Two participants were excluded because

they attended only one class. Two participants were excluded because they were the only attendees in their classes. The final sample included 46 participants, with a mean age of 18.98 years ($SD \pm 1.48$), a mean height of 1.66 m ($SD \pm 0.07$), a mean weight of 64.48 kg ($SD \pm 10.89$), and a mean body mass index of 23.6 ($SD \pm 3.6$). On average, participants exercised 1.2 ($SD \pm 0.62$) times per week. Trait social physique anxiety was 2.99 ($SD \pm 0.67$), typical of female university samples.

Materials

Demographics. Participants self-reported age, height, weight, major and current physical activity behavior (average number of days per week engaged in exercise in the past 6 months).

Physical Activity Clearance. Participants completed the Physical Activity Readiness Questionnaire (PAR-Q³⁰), a 7-item measure that assesses whether an individual can perform physical activity safely. Participants answer yes or no to each item. All participants responded “no” to all items and were cleared for physical activity.

Body Satisfaction. The Body Image State Scale (BISS³¹) was used to assess state body satisfaction. It consists of 6 items examining dissatisfaction/satisfaction with one's overall physical appearance, body shape, size, weight, and physical attractiveness. Each item is rated on a 9-point Likert scale, with higher scores representing higher body satisfaction. Evidence of reliability and validity of the BISS has been shown in previous research.³¹ Reliability for preclass and postclass BISS scores in the current study were adequate (Cronbach α range = 0.82–0.86).

State Social Physique Anxiety. The State Social Physique Anxiety Scale (SPAS-S²³) was used to assess concern over the body being evaluated by others. Participants rate each of the 9 items on a 5-point Likert scale ranging from 1 (*not at all characteristic of me*) to 5 (*extremely characteristic of me*). Higher scores represent greater social physique anxiety. Evidence of reliability and validity of the SPAS-S has been shown in previous research.^{23,32} Reliability for preclass and postclass SPAS-S scores in the current study were adequate (Cronbach α range = 0.91–0.92).

Trait Social Physique Anxiety. The 9-item version of the Social Physique Anxiety Scale³³ was used to assess concern over one's body being evaluated by others. This measure has shown improved validity and reliability over the original 12-item version.²⁴ Each item is rated on a scale ranging from 1 (*not at all characteristic of me*) to 5 (*completely characteristic of me*), and higher scores represent higher levels of concern. Internal consistency reliability was adequate (Cronbach α = 0.70).

Design

A pretest-posttest repeated-measures design was used. Each participant completed both types of exercise classes with the same group of participants. Participants completed state body image measures immediately before and immediately after each class. Classes were separated by 1 week, with class order counterbalanced (ie, half the groups completed the yoga class first and half completed the resistance class first) to eliminate order effects.

Procedures

Upon arrival at the laboratory for the first session, participants provided informed consent and demographic information and

completed the PAR-Q. Next, with the assistance of the exercise instructor, they gathered the equipment required for the particular class (ie, a mat for the yoga class, a mat and a set of 3- to 5-lb hand weight for the resistance class). The participants found a place on the exercise floor where they completed the preexercise state body image measures. The exercise class followed (see below for descriptions), after which participants completed the same body image measures. All measures (preexercise and postexercise) were counterbalanced to avoid potential order effects. At the second session, participants followed the same basic protocol; however, they completed the other class type. Upon completion of the second session, the participants were debriefed as to the true purpose of the study.

Exercise Classes. Each of the classes was designed by the first two authors, both certified fitness instructors through Canadian Fitness Professionals and the YMCA. They were designed to be suitable for beginner exercisers working at a moderate intensity level. Classes had 6 to 8 participants. The classes were approximately 30 minutes in length, each consisting of a 5-minute warm-up, 20 minutes of weight training or yoga, and a 5-minute cool-down and stretch. Feedback from the instructor was limited to instructional cues only. The physical environment and exercise leader were identical for both classes. Specifically, classes took place in an open group-exercise space that had mirrors on 2 walls. In addition, there was strength training and cardiovascular exercise equipment in the remainder of the space, as is typical in an exercise facility. Although some research has shown that riding a stationary bike or running on a treadmill in front of a mirror may lead to more negative body image,³⁴ other research has shown that group exercise classes (ie, resistance training) in spaces with mirrors do not negatively affect body image in similar samples.¹⁶

The yoga class consisted of introductory Hatha-style yoga, which involved participants moving through a series of poses designed to improve balance, muscle strength, and flexibility. Throughout the class, deep breathing was linked to each movement. All poses were suitable for beginners (eg, child's pose, downward dog). The 5-minute warm-up included an explanation of the breathing technique; as is typical in a yoga class, a 5-minute final relaxation was included as the cool-down. Yoga was chosen for 3 reasons. First, the growing popularity of yoga⁵ as a form of exercise suggests that more research on its benefits is needed. Second, correlational evidence^{7,10} in women shows yoga is associated with improved body image (eg, self-objectification) compared with either resistance training or cardiovascular exercise. Thus, there was reason to believe yoga may be more effective. Third, yoga has shown numerous psychological benefits, including reduced stress and anxiety;³⁵⁻³⁸ thus it is possible it could be associated with improvements in body image.

The resistance class was modeled after that in a previous study¹⁵ showing improvements in social physique anxiety after the class. It used free-weight and body-weight exercises to improve muscular strength and endurance. The weight training exercises targeted each major muscle group with one exercise (eg, bicep curls, chest press, squats). One set of 12 repetitions (or as many as could be performed correctly) of each exercise was completed. Finally, a 5-minute cool-down and stretching component was conducted at the end of the class, covering all the major muscle groups that were used during the class. A detailed class outline is available from the corresponding author.

Instructor Characteristics and Qualifications. The second author conducted all classes. At the time of the study, she was a certified Personal Training Specialist, Group Fitness Instructor Specialist, and indoor group cycling instructor through Canadian Fitness Professionals. She had 4 years of experience instructing a variety of group fitness classes and 2 years as a regular yoga practitioner.

Instructor Training for the Current Study. The instructor practiced leading each class several times to ensure that the exercise order, intensity level, duration, and overall class design were effective and appropriate for nonexercisers or infrequent exercisers. The classes were practiced with a group of individuals who had fitness and experience levels similar to those of the potential participants (but who were excluded from participating in the study) and in front of the first author until consistency of instruction was reached. A checklist was created for this class and was used during these practice sessions classes to ensure the consistency of instruction in all classes.

Class Consistency Manipulation Check. During each class, out of view of the participants, a research assistant used a checklist to assess class consistency, which outlined in detail the order of the exercises to be performed, the number of repetitions to be performed, and the instructional cues to be provided. The researcher recorded each error (ie, missed exercise, exercise out of sequence, missed cue, or additional cue) as well as the starting and finishing times of each class. For each class, the percentage of correct exercises and instructions was calculated.

Statistical Analysis

To compare the effects of single yoga and resistance sessions on state body satisfaction and social physique anxiety, a 2 (time: preexercise and postexercise) $\times$ 2 (class type: yoga, resistance) repeated-measures multiple analysis of variance (MANOVA) was conducted, with repeated measures on both factors. A multivariate approach was used because it does not require the assumption of sphericity. Follow-up univariate analyses of variance (ANOVAs) were used to investigate significant multivariate effects. Paired *t*-tests were used to follow-up significant ANOVAs.

Results

Data were initially screened to ensure variables were normally distributed and that assumptions for statistical analyses were met, following recommendations.³⁹ Descriptive statistics for preexercise and postexercise body satisfaction and social physique anxiety scores by class type are shown in Table 1. Scores on the BISS⁴⁰⁻⁴² and the S-SPAS^{15,16} before the exercise classes were typical of other samples of university women. Paired *t* tests comparing preexercise body satisfaction and social physique anxiety between yoga and resistance classes showed no significant differences before the exercise classes (*P* values $> .05$). The checklist for consistency ranged from 89% to 100% of the correct exercises and cues provided for each class taught.

Results of the MANOVA showed there was a significant main effect of time ($F_{2,44} = 8.58$, $P < .05$, $\eta_p^2 = 0.28$) but no main effect of class type ($P > .05$). The time effect was superseded by a significant overall Time $\times$ Class Type interaction ($F_{2,44} = 5.69$, $P < .01$, $\eta_p^2 =$

Table 1 Preexercise and Postexercise Means (Standard Deviations) for Body Satisfaction and Social Physique Anxiety Scores by Class Type

Variable	Yoga		Resistance	
	Preexercise	Postexercise	Preexercise	Postexercise
State body satisfaction (1–9)	5.05 (1.45)	5.50 (1.44)	5.10 (1.44)	5.15 (1.55)
State social physique anxiety (1–5)	2.85 (.94)	2.45 (.91)	2.84 (.94)	2.71 (.99)

Note. Higher scores represent higher body satisfaction or higher anxiety.

0.21). Follow-up ANOVAs showed significant effects on both body satisfaction ($F_{1,45} = 7.48, P < .01, \eta_p^2 = .14$) and social physique anxiety ($F_{1,45} = 5.46, P < .05, \eta_p^2 = 0.11$).

To examine the nature of the significant interactions, we conducted 4 paired *t* tests to examine the change in body satisfaction and social physique anxiety over time, within each class type. For the yoga class, there was a significant difference for both body satisfaction ($t_{45} = -3.76, P = .001$, Cohen's $d^{43} = 0.52$) and social physique anxiety ($t_{45} = 4.57, P < .001$, Cohen's $d = 0.67$) with a significant increase in body satisfaction and a significant decrease in social physique anxiety. For the resistance class, there was a significant decrease in social physique anxiety ($t_{45} = 2.15, P = .022$, Cohen's $d = 0.31$; see Figure 1 for all interactions). In addition, a paired *t* test indicated that body satisfaction ($t_{46} = -2.22, P = .032$) was higher and social physique anxiety ($t_{46} = 2.86, P = .006$) was lower after the yoga class versus the resistance class.

Discussion

The current study examined whether a single yoga or resistance group exercise class would lead to improvements in state body image in female college nonexercisers or infrequent exercisers. The hypotheses were generally supported; yoga was associated with increases in body satisfaction and decreases in social physique anxiety, whereas the resistance class led to decreases in social physique anxiety. Effect sizes showed that the magnitude of the changes for both classes was small-to-moderate (consistent with meta-analytic findings²) and was larger after the yoga class than after the resistance class.

Our findings showed that significant improvements in body satisfaction were found only after the yoga class, consistent with correlational research showing that yoga participation is associated with greater body satisfaction compared with other types of exercise or other treatments.^{7,9,44} The current study extends this research by showing state body satisfaction can be improved after a single yoga session.

Yoga may lead to improved body satisfaction through increased body awareness and responsiveness (termed embodiment^{7,8,45,46}). For example, Daubenmier⁷ found that in yoga participants, body awareness and responsiveness mediated improved body satisfaction such that being more attentive to internal body sensations led to greater satisfaction with the body. Thompson and colleagues⁴⁷ suggested that a lack of body awareness is one of several factors related to the development of body dissatisfaction, because individuals are unable to recognize and respond to their body's sensations and cues effectively. Consistent with this suggestion, Clance and colleagues^{44,48} found that body awareness training and yoga training

were both associated with increases in body satisfaction in college students and children.

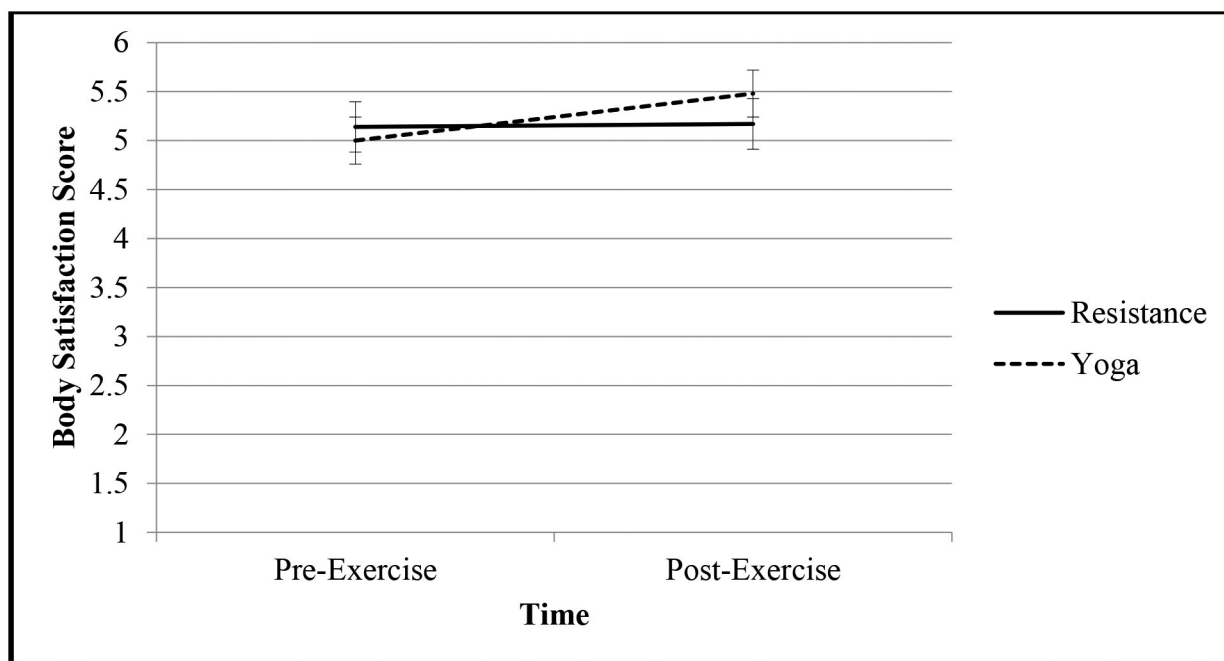
Not surprisingly, our results showed that a single session of group exercise (yoga or resistance) reduced state social physique anxiety, consistent with previous research in similar samples.^{15,16} These findings reinforce the powerful influence of exercise (even a single session) on reducing negative affect, but extend this research to other types of exercise (ie, yoga).

The mechanisms attributed to this change can only be speculated upon in the current study. It is possible that successful completion of both the yoga and resistance classes led to increases in self-efficacy, which may help to reduce anxiety.⁴⁹ It is also possible that the focus on internal sensations of the body (for the yoga class) or the body's function (for the resistance class) versus physical appearance may lead to reductions in negative state body image. Specifically, during yoga, participants are encouraged to focus on their body sensations; thus, participants may be less inclined to focus on how others perceive their body. For the resistance class, which focuses on technique and body alignment (rather than appearance), participants may tend to focus less on their physical appearance and more on what their bodies can do (ie, function or fitness). If women do not focus on their appearance, they may in turn worry less about what others think about their appearance.

Upon comparing the effects of the yoga class and the resistance class, we found that the magnitude of changes for body image were larger after the yoga class. The reason for yoga's greater effectiveness may be the shift in attention from physical appearance to embodiment. Although there can be a shift in attention from physical appearance to form and technique in a resistance class, it still involves an outward monitoring of the body, perhaps limiting the magnitude of the change in negative body image and having no effect on body satisfaction. Alternatively, the differing motives for resistance training versus yoga may be influential. For example, although resistance training is performed to increase muscular strength and endurance or increase muscle tone, the 2 most common reasons for participating in yoga are flexibility and stress relief, which are unlikely to directly lead to changes in appearance.

However, aside from focusing less on the outward appearance of the body, the greater magnitude of changes found for the yoga class may be due to the philosophy and instruction provided during yoga. Yoga emphasizes the mind-body connection through linking breath to movement and asking participants to simply experience the sensations in the body, without trying to change them. The emphasis on deep breathing and clearing the mind from outside distractions could help reduce the cognitive and somatic symptoms of anxiety, which could contribute to a reduction in state social physique anxiety and promote an overall sense of well-being. It may also be that the concentration and effort required to listen to the body while

(a)



(b)

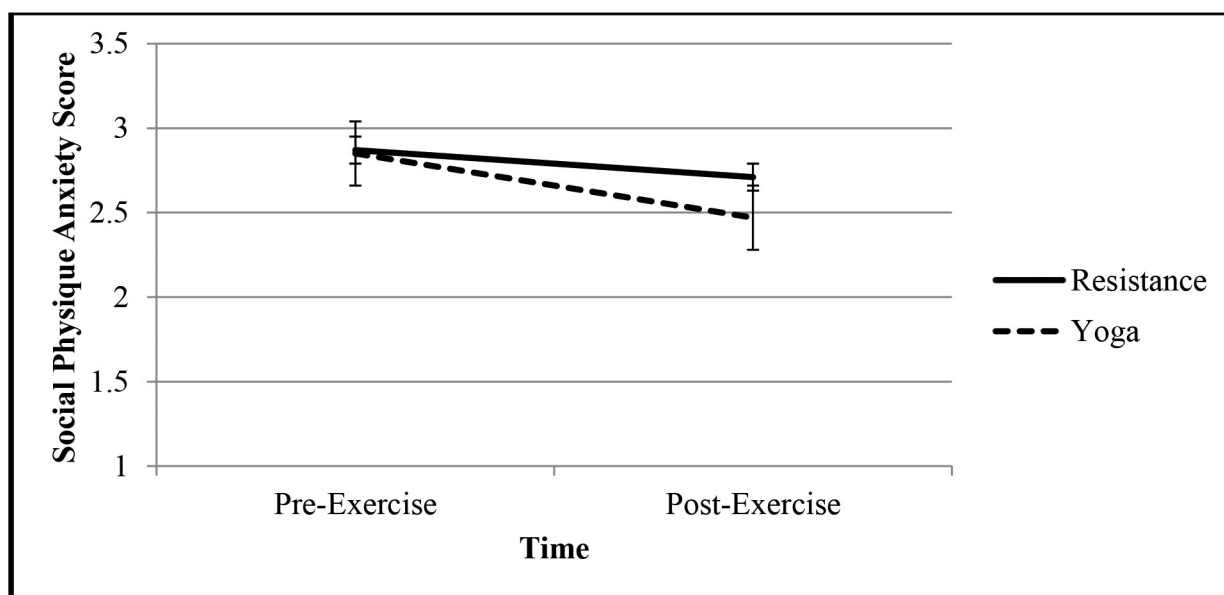


Figure 1 — **a.** Relationship between body satisfaction score and time, presented separately by class type. There was a significant increase in body satisfaction after the yoga class, but no change in body satisfaction after the resistance class. **b.** Relationship between social physique anxiety score and time, presented separately by class type. There was a significant decrease in social physique anxiety after the yoga and resistance classes, but a larger decrease after the yoga class.

maintaining a pose and breathing simply does not allow participants to pay attention to their appearance or what others are thinking.

Further, there is a focus on relaxation in yoga. Evidence suggests that yoga can help reduce symptoms of anxiety, including heart rate and blood pressure, in as little as a single session.^{35–38} In Hatha yoga specifically, participants are encouraged to work at their own pace and level, to avoid comparison with others or judgment of themselves,⁵⁰ and to pay attention to how the body feels, rather than external cues (from an instructor, music, or their appearance⁴⁴). Thus, feeling more relaxed after class may help to reduce body image concerns, and specifically social physique anxiety, after yoga.

Another possibility for the greater effects of a yoga class compared with a resistance class may be related to other variables not measured in the current study. For example, studies have shown yoga participation to be associated with less self-objectification,^{7,8,10,44} which is associated with decreased self-consciousness.⁵¹ Lower self-consciousness is associated with lower social physique anxiety.^{24,52} Compared with a resistance class, a yoga class may be structured such that self-objectification and self-consciousness are minimized, thus reducing negative body image.

Limitations

Some limitations should be acknowledged. We examined one specific type of yoga class, but there are numerous variations, and they could have differing effects. We also examined only 2 specific body image-related outcomes. Other body image variables (ie, self-objectification, body shame) may change differently in response to yoga. Second, although yoga is increasing in popularity, it is likely that participants were less familiar with poses in the yoga class than exercises in the resistance class. Third, we were interested in the effects of a single class. Although the classes were separated by a week, it is possible that there could have been carryover effects from the first class to the second. We did counterbalance the order of the classes (ie, half the groups did the yoga class first and half did the resistance class first) to avoid this influence, and there were no differences in any of the body image variables between the 2 types of classes at preexercise. Although we attempted to replicate a “real-life” fitness setting, the classes were all held in a laboratory, thereby limiting ecological validity. Finally, it should be noted that as follow-ups, we did multiple *t* tests, potentially inflating the type I error rate.

Implications and Future Research

There are several practical implications of the current study. First, the present findings further reinforce the effectiveness of exercise as a method of improving body image^{2–4} but extend that to a single exercise session. From a practical standpoint, it is more feasible to encourage people to commit to participating in one single exercise session, rather than commit to a long-term exercise program. When exercisers experience more positive affective states, they are more likely to adhere to exercise.⁵³ Given that yoga was more effective in improving body image than a resistance class, it may be particularly relevant to promote mind-body activities to maximize benefits. Alternatively, it may be useful to incorporate elements of yoga into more traditional types of fitness activities to allow participants to experience similar body image benefits in other types of exercise. For example, previous research has shown that improved body awareness is associated with less self-objectification¹⁰; although we can only speculate, focusing on body awareness and responsiveness,

deep breathing, or relaxation may lead to more positive body image outcomes in any type of exercise class.

Future research should replicate the findings of the current study in other samples, such as those at risk for negative body image (eg, individuals with an eating disorder, those who are overweight, or adolescent girls). In addition, it would be interesting to determine whether similar effects would be found in samples of older women or men, groups with different body image concerns than university women. Further, it is important to examine the effects of a long-term yoga training program on trait body image; evidence suggests that greater time spent practicing yoga is associated with larger improvements in body image.^{7,10} Finally, the time course of these changes (ie, how much training is required to see changes to trait body image) and issues related to dose response should also be investigated.

Conclusions

The current study examined whether a single yoga class could improve state body image, and if so, compare those changes with those from a resistance class. The findings were encouraging; both types of classes were associated with body image improvements, and greater improvements followed the yoga class. This study extends the correlational evidence suggesting that yoga may be particularly beneficial for improving body image in college women. Given that negative body image is associated with increased negative mental health outcomes including depression and eating disorders,⁵⁴ it is important to continue to increase our understanding of ways to decrease the negative body image that is common in young adult women. It also suggests that research should continue to investigate how exercise can be structured to maximize the psychological benefits that individuals receive.

References

1. Cash TF, Henry PE. Women's body images: the results of a national survey in the U.S.A. *Sex Roles*. 1995;33:19–28. doi:10.1007/BF01547933
2. Campbell A, Hausenblas HE. Effects of exercise interventions on body image: a meta-analysis. *J Health Psychol*. 2009;14:780–793. PubMed doi:10.1177/1359105309338977
3. Hausenblas HA, Fallon EA. Exercise and body image: a meta-analysis. *Psychol Health*. 2006;21:33–47. doi:10.1080/14768320500105270
4. Reel JJ, Greenleaf C, Baker WK, et al. Relations of body concerns and exercise behaviour: a meta-analysis. *Psychol Rep*. 2007;101:927–942. PubMed
5. Yoga Journal and Yoga Alliance. *Yoga in America study*. <http://media.yogajournal.com/wp-content/uploads/2016-Yoga-in-America-Study-Comprehensive-RESULTS.pdf>. Published 2016. Accessed May 8, 2016.
6. Mayo Clinic. *Yoga: Fight stress and find serenity*. <http://www.mayoclinic.org/healthy-lifestyle/stress-management/in-depth/yoga/art-20044733>. Published 2015. Accessed November 16, 2016.
7. Daubenmier JJ. The relationship of yoga, body awareness, and body responsiveness to self-objectification and disordered eating. *Psychol Women Q*. 2005;29:207–219. doi:10.1111/j.1471-6402.2005.00183.x
8. Dittmann KA, Freedman MR. Body awareness, eating attitudes, and spiritual beliefs of women practicing yoga. *Eat Disord*. 2009;17:273–292. PubMed doi:10.1080/10640260902991111

9. Neumark-Sztainer D, Eisenberg ME, Wall M, Loth KA. Yoga and pilates: associations with body image and disordered-eating behaviors in a population-based sample of young adults. *Int J Eat Disord*. 2011;44:276–280. [PubMed doi:10.1002/eat.20858](#)
10. Prichard I, Tiggemann M. Relations among exercise type, self-objectification, and body image in the fitness center environment: the role of reasons for exercise. *Psychol Sport Exerc*. 2008;9:855–866. [doi:10.1016/j.psychsport.2007.10.005](#)
11. Boudette R. Question & answer: yoga in the treatment of disordered eating and body image disturbance. How can the practice of yoga be helpful in recovery from and eating disorder? *Eat Disord*. 2006;14:167–170. [PubMed doi:10.1080/10640260500536334](#)
12. Mitchell KS, Mazzeo SE, Rausch SM, Cooke KL. Innovation interventions for disordered eating: evaluating dissonance-based and yoga interventions. *Int J Eat Disord*. 2007;40:120–128. [PubMed doi:10.1002/eat.20282](#)
13. Scime M, Cook-Cottone C. Primary prevention of eating disorders: a constructivist integration of mind and body strategies. *Int J Eat Disord*. 2008;41:134–142. [PubMed doi:10.1002/eat.20480](#)
14. Scime M, Cook-Cottone C, Kane L, Watson T. Group prevention of eating disorders with fifth-grade females: IMPACT on body dissatisfaction, drive for thinness, and media influence. *Eat Disord*. 2006;14:143–155. [PubMed doi:10.1080/10640260500403881](#)
15. Lamarche L, Gammage KL. The effects of leader gender on self-presentational concerns in exercise. *Psychol Health*. 2010;25:769–781. [PubMed doi:10.1080/08870440902866886](#)
16. Lamarche L, Gammage KL, Strong HA. The effect of mirrored environments on self-presentational efficacy and social anxiety in women in a step aerobics class. *Psychol Sport Exerc*. 2009;10:67–71. [doi:10.1016/j.psychsport.2008.06.006](#)
17. McInman AD, Berger BG. Self-concept and mood changes associated with aerobic dance. *Aust J Psychol*. 1993;45:134–140. [doi:10.1080/00049539308259130](#)
18. Dishman RK, Sallis JF, Orenstein DR. The determinants of physical activity and exercise. *Public Health Rep*. 1985;45: 158–171.
19. Dishman RK. Determinants of participation in physical activity. In: Bouchard C, Shephard RJ, Stephens T, et al, eds. *Exercise, Fitness, and Health: A Consensus of Current Knowledge*. Champaign, IL: Human Kinetics; 1990:75–101.
20. Ekkekakis P, Petruzzello SJ. Acute aerobic exercise and affect: current status, problems, and prospects regarding dose-response. *Sports Med*. 1999;28:337–374. [PubMed doi:10.2165/00007256-199928050-00005](#)
21. Heatherton TF, Mahamedi F, Striepe M, Field AE, Keel PK. A 10-year longitudinal study of body weight, dieting, and eating disorder symptoms. *J Abnorm Psychol*. 1997;106:117–125. [PubMed doi:10.1037/0021-843X.106.1.117](#)
22. Striegel-Moore RH, Franko DL. Body image issues among girls and women. In: Cash TF, Pruzinsky T, eds. *Body Image: A Handbook of Theory, Research, & Clinical Practice*. New York: The Guildford Press; 2002:183–191.
23. Kruisselbrink LD, Dodge AM, Swanburg SL, MacLeod AL. Influence of same-sex and mixed sex exercise settings on the social physique anxiety and exercise intentions of males and females. *J Sport Exer Psychol*. 2004;26:616–622. [doi:10.1123/jsep.26.4.616](#)
24. Hart EA, Leary MR, Rejeski WJ. The measurement of social physique anxiety. *J Sport Exer Psychol*. 1989;11:94–104. [doi:10.1123/jsep.11.1.94](#)
25. Fleming JC, Martin Ginis KA. The effects of commercial exercise video models on women's self-presentational efficacy and exercise task self-efficacy. *J Appl Sport Psychol*. 2004;16:92–102. [doi:10.1080/10413200490260080](#)
26. Gammage KL, Hall CR, Martin Ginis KA. Self-presentation in exercise contexts: differences between high and low frequency exercisers. *J Appl Soc Psychol*. 2004;34:1638–1651. [doi:10.1111/j.1559-1816.2004.tb02791.x](#)
27. Martin Ginis KA, Eng JJ, Arbour KP, Hartman JW, Phillips SM. Mind over muscle? Sex differences in the relationship between body image change and subjective and objective physical changes following a 12-week strength-training program. *Body Image*. 2005;2:363–372. [PubMed doi:10.1016/j.bodyim.2005.08.003](#)
28. Rodgers WM, Gauvin L. Heterogeneity of incentives for physical activity and self-efficacy in high active and moderately active women exercisers. *J Appl Soc Psychol*. 1998;28:1016–1029. [doi:10.1111/j.1559-1816.1998.tb01665.x](#)
29. Ginis KA, Prapavessis H, Haase AM. The effects of physique-salient and physique non-salient exercise videos on women's body image, self-presentational concerns, and exercise motivation. *Body Image*. 2008;5:164–172. [PubMed doi:10.1016/j.bodyim.2007.11.005](#)
30. Canadian Society for Exercise Physiology. *Physical Activity Readiness Questionnaire (PAR-Q)*. <http://www.csep.ca/CMFiles/publications/parq/par-q.pdf>. Published 2002. Updated July 14, 2014. Accessed August 17, 2012.
31. Cash TF, Fleming EC, Alindogan J, Steadman L, Whitehead A. Beyond body image as a trait: the development and validation of the body image states scale. *Eat Disord*. 2002;10:103–113. [PubMed doi:10.1080/10640260290081678](#)
32. Martin Ginis KA, Murru E, Conlin C, Strong HA. Construct validation of a state version of the social physique anxiety scale among young women. *Body Image*. 2011;8:52–57. [PubMed doi:10.1016/j.bodyim.2010.10.001](#)
33. Martin KA, Rejeski WJ, Leary MR, McAuley E, Bane S. Is the Social Physique Anxiety Scale really multidimensional? Conceptual and statistical arguments for a unidimensional model. *J Sport Exer Psychol*. 1997;19:359–367. [doi:10.1123/jsep.19.4.359](#)
34. Martin Ginis KA, Jung ME, Gauvin L. To see or not to see: Effects of exercising in mirrored environments on sedentary women's feeling states and self-efficacy. *Health Psychol*. 2003;22:354–361. [PubMed doi:10.1037/0278-6133.22.4.354](#)
35. Berger BG, Owen DR. Stress reduction and mood enhancement in four exercise modes: swimming, body conditioning, Hatha yoga, and fencing. *Res Q Exerc Sport*. 1988;59:148–159. [doi:10.1080/02701367.1988.10605493](#)
36. Cowen VS, Adams TB. Physical and perceptual benefits of yoga asana practice. *J Bodyw Mov Ther*. 2005;9:211–219. [doi:10.1016/j.jbmt.2004.08.001](#)
37. Cusumano JA, Robinson SE. The short-term psychophysiological effects of Hatha yoga and progressive relaxation on female Japanese students. *App Psychol Int Rev*. 1992;42:77–90. [doi:10.1111/j.1464-0597.1993.tb00725.x](#)
38. Netz Y, Lindor R. Mood alterations in mindful versus aerobic exercise modes. *J Psychol*. 2003;137:405–419. [PubMed doi:10.1080/00223980309600624](#)
39. Tabachnick BG, Fidell LS. *Using Multivariate Statistics*. 5th ed. Toronto, ON, Canada: Pearson; 2007.
40. Rudiger JA, Cash TF, Roehrig M, Thompson JK. Day-to-day body-image states: prospective predictors of intra-individual level and variability. *Body Image*. 2007;4:1–9. [PubMed doi:10.1016/j.bodyim.2006.11.004](#)
41. Vocks S, Hechler T, Rohrig S, Legenbauer T. Effects of a physical exercise session on state body image: the influence of pre-experimental body dissatisfaction and concerns about weight and shape. *Psychol Health*. 2009;24:713–728. [PubMed doi:10.1080/08870440801998988](#)

42. Yamamiya Y, Cash TF, Melnyk SE, Posavac HD, Posavac SS. Women's exposure to thin-and-beautiful media images: body image effects of media-ideal internalization and impact-reduction interventions. *Body Image*. 2005;2:74–80. [PubMed doi:10.1016/j.bodyim.2004.11.001](#)
43. Cohen J. A power primer. *Psychol Bull*. 1992;112:155–159. [PubMed doi:10.1037/0033-2909.112.1.155](#)
44. Clance PR, Mitchell M, Engelman SR. Body cathexis in children as a function of awareness training and yoga. *J Clin Child Psychol*. 1980;9:82–85. [doi:10.1080/15374418009532956](#)
45. Impett EA, Daubenmier JJ, Hirschman AL. Minding the body: Yoga, embodiment, and well-being. *Sex Res Soc Policy*. 2006;3:39–48. [doi:10.1525/srsp.2006.3.4.39](#)
46. Rani NJ, Rao PV. Body awareness and yoga training. *Percept Mot Skills*. 1994;79:1103–1106. [doi:10.2466/pms.1994.79.3.1103](#)
47. Thompson JK, Heinberg LJ, Altabe M, Tantleff-Dunn S. *Exacting Beauty: Theory, Assessment, and Treatment of Body Image Disturbance*. Washington, DC: American Psychological Association; 1999. [doi:10.1037/10312-000](#)
48. Glance PR, Matthews TV, Joesting J. Body-cathexis and self-cathexis in an interactional, awareness training class. *Percept Mot Skills*. 1979;48:221–222. [PubMed doi:10.2466/pms.1979.48.1.221](#)
49. Petruzzello SJ, Landers DM, Hatfield BD, Kubitz KA, Salazar W. A meta-analysis on the anxiety-reducing effects of acute and chronic exercise: outcomes and mechanisms. *Sports Med*. 1991;11:143–182. [PubMed doi:10.2165/00007256-199111030-00002](#)
50. Shaw B. *Yogafit*. 2nd ed. Champaign, IL: Human Kinetics; 2008.
51. Fredrickson BL, Roberts T, Noll SM, Quinn DM, Twenge JM. The swimsuit becomes you: sex differences in self-objectification, restrained eating, and math performance. *J Pers Soc Psychol*. 1998;75:269–284. [PubMed doi:10.1037/0022-3514.75.1.269](#)
52. Sabiston CM, Sedgwick WA, Crocker PRE, Kowalski KC, Mack DE. Social physique anxiety in adolescence: an exploration of influences, coping strategies, and health behaviours. *J Adolesc Res*. 2007;22:78–101. [doi:10.1177/0743558406294628](#)
53. Williams DM, Dunsiger S, Ciccolo JT, Lewis BA, Albrecht AE, Marcus BH. Acute affective response to a moderate-intensity exercise stimulus predicts physical activity participation 6 and 12 months later. *Psychol Sport Exerc*. 2008;9:231–245. [PubMed doi:10.1016/j.psychsport.2007.04.002](#)
54. Fredrickson BL, Roberts TA. Objectification theory. *Psychol Women Q*. 1997;21:173–206. [doi:10.1111/j.1471-6402.1997.tb00108.x](#)